ACML101

Physical Chemistry Assignment 2

Instructor

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Fill in the blank

1. The Normalization constant (N) for $\psi = N sin\frac{n\pi x}{L}$ in the range $0 < x < L$ is ……… $\sqrt{2/L}$

2. In polar coordinates, $\hat{L}^2$ is given by

$$\left[-\hbar^2\left(\frac{1}{sin\theta}\frac{\partial}{\partial\theta}sin\theta\frac{\partial}{\partial\theta}+\frac{1}{sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right)\right]$$

Eigen value of the function $(3cos^2\theta - 1)$ for the operator $\hat{L}^2$ is ………… $6\hbar^2$

3. Consider the following reaction

$A + B \rightleftharpoons C + D$ (Take rate constant for forward and backward reaction as k_1 and k_{-1})

$C \rightarrow P$ (Take rate constant as k_2)

$\frac{dP}{dt}$ in terms of reactants and products is ………… $k_2 \frac{k_1[A][B]}{k_{-1}[D]+k_2}$

4. The de Broglie wavelength of an electron moving with a speed of 2.0×10^6 m/s is

(mass of electron= 9.1×10^{-31} kg) 3.64×10^{-10} m , 3.64 A°, 0.364 nm

True/False

5. coskx is an eigen function of operator d/dx. False

6. $-i\hbar\frac{\partial}{\partial x}$ is a linear operator. TRUE

7. The operator $\hat{p}_x$ commutes with operator $\hat{x}$. FALSE

8. In the Haber's process of synthesis of ammonia, temperature is kept higher since the reaction is exothermic. FALSE